

Yelp Data Cleaning

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#Loading

#Part 1: Files of Philadelphia Created

1.1 Get PH Business data

```
business_file <- paste0(data_dir0, "/1.yelp_academic_dataset_business.json")

business <- stream_in(file(business_file), verbose = FALSE)

# Filter Philadelphia
philly_business <- filter(business, city == "Philadelphia")
dim(philly_business) #14569    14

# Filter categories
categories <- philly_business$categories |>
  strsplit(",") |>
  unlist() |>
  trimws() |> #cut spaces
  table()

popular_categories <- as.data.frame(categories) |>
  rename(Category = Var1) |> arrange(desc(Freq))

# Filter restaurants
restaurant_keywords <- c("restaurants", "food", "bars")
```

```

is_restaurant <- function(categories) {
  if(is.na(categories)) return(FALSE)
  categories_lower <- tolower(categories)
  any(sapply(restaurant_keywords, function(kw) grepl(kw, categories_lower)))
}

# Add "price_level"
#table(philly_restaurants$attributes$RestaurantsPriceRange2)
#1      2      3      4 None
#2731 3093  338   42    2

philly_restaurants <- philly_business |>
  rowwise() |>
  filter(is_restaurant(categories)) |>
  filter(!is.na(attributes$RestaurantsPriceRange2)) |>
  filter(attributes$RestaurantsPriceRange2 != "None") %>%
  mutate(
    price_level = as.numeric(attributes$RestaurantsPriceRange2)
  ) |>
  ungroup()

cat("There're valid Philadelphia restaurants are:\n", nrow(philly_restaurants))

# Save PH Restaurants Data (0324)
philly_restaurants2 <- philly_restaurants |> select(-c(address, postal_code, latitude, longitude))
dim(philly_restaurants2) #6204
head(philly_restaurants2)
#write_csv(philly_restaurants2, paste0(data_dir, 'philly_restaurants.csv'))

```

1.2.1 Get PH Review Data

```

# Filter reviews of Philly restaurants
philly_rest_ids <- unique(philly_restaurants$business_id)

review_file <- paste0(data_dir0, "/2.yelp_academic_dataset_review.json")
philly_review <- "philly_reviews.json"

con_in <- file(review_file, open = "r")

```

```

con_out <- file(philly_review, open = "w")

stream_in(
  con_in,
  handler = function(df) {
    matching <- filter(df, business_id %in% philly_rest_ids)
    if (nrow(matching) > 0) {
      stream_out(matching, con_out, pagesize = nrow(matching), verbose = FALSE)
    }
  },
  pagesize = 10000
)

close(con_in)
close(con_out)

# Load
philly_reviews <- stream_in(file(philly_review))
nrow(philly_reviews)#730509

head(philly_reviews)

#save JSON
#write_lines(
  # toJSON(philly_reviews, auto_unbox = TRUE),
  #"philly_reviews_data.json")
#write_json(philly_reviews, "philly_reviews_data.json")

```

1.2.2 Get PH tips data

```

#user
#philly_review_user_ids <- unique(philly_reviews$user_id)
philly_restaurants<-read.csv( "/Users/a1-6/Desktop/CSS/1.Thesis/Data/philly_restaurants.csv")
dim(philly_restaurants)#6204

# Filter users to those who reviewed Philly restaurants
tip_file <- paste0(data_dir0, "/yelp_academic_dataset_tip.json")

philly_tip <- "philly_tips.json"

```

```

con_in <- file(tip_file, open = "r")
con_out <- file(philly_tip, open = "w")

stream_in(
  con_in,
  handler = function(df) {
    matching <- filter(df, business_id %in% philly_restaurants$business_id)
    if (nrow(matching) > 0) {
      stream_out(matching, con_out, pagesize = nrow(matching), verbose = FALSE)
    }
  },
  pagesize = 10000
)

close(con_in)
close(con_out)

# load
philly_tips <- stream_in(file(philly_tip), verbose = FALSE)
cat("There're:", nrow(philly_tips), "users who visits the restaurants. \n")

head(philly_tips)#95763

#save JSON
#write_lines(
  # toJSON(philly_tips, auto_unbox = TRUE),
  # "philly_tips_data.json" )

```

1.2.3 merge PH review data

```

philly_reviews <- stream_in(file("philly_reviews_data.json"))

philly_reviews <- philly_reviews |> select(-c(useful,funny,cool))
colnames(philly_reviews)
colnames(philly_tips)

# Save PH Review Data
all_philly_reviews <- bind_rows(philly_reviews, philly_tips) |> select(-c(review_id,compliment_count))
colnames(all_philly_reviews)
dim(all_philly_reviews) #826272

```

```
#write_csv(all_philly_reviews, paste0(data_dir, 'all_philly_reviews.csv')
```

1.3 Get PH user data

```
#user
philly_review_user_ids <- unique(all_philly_reviews$user_id)

# Filter users to those who reviewed Philly restaurants
user_file <- paste0(data_dir0, "/3.yelp_academic_dataset_user.json")

philly_user <- "philly_users.json"

con_in <- file(user_file, open = "r")
con_out <- file(philly_user, open = "w")

stream_in(
  con_in,
  handler = function(df) {
    matching <- filter(df, user_id %in% philly_review_user_ids)
    if (nrow(matching) > 0) {
      stream_out(matching, con_out, pagesize = nrow(matching), verbose = FALSE)
    }
  },
  pagesize = 10000
)

close(con_in)
close(con_out)

# load
philly_users <- stream_in(file(philly_user), verbose = FALSE)
cat("There're:", nrow(philly_users), "users who visits the restaurants. \n")

dim(philly_users)#220404
#save JSON
#write_lines(
  # toJSON(philly_users, auto_unbox = TRUE),
  #"philly_users.json")
```

Part 2: Get friends Business Records

```
omni_friend_df <- read.csv(paste0(data_dir, 'Omnivore_friend_df0319.csv'))

dim(omni_friend_df) #43632
head(omni_friend_df)

friends_ids <- omni_friend_df$user_id
review_file <- paste0(data_dir0, "/2.yelp_academic_dataset_review.json")

batch_size <- 10000
friend_batches <- split(friends_ids, ceiling(seq_along(friends_ids) / batch_size))

all_friend_reviews <- list()

for (i in seq_along(friend_batches)) {

  batch_ids <- friend_batches[[i]]

  con_in <- file(review_file, open = "r")
  batch_file <- tempfile()
  con_out <- file(batch_file, open = "w")

  stream_in(
    con_in,
    handler = function(df) {
      matching <- filter(df, user_id %in% batch_ids)
      if (nrow(matching) > 0) {
        stream_out(matching, con_out, pagesize = nrow(matching), verbose = FALSE)
      }
    },
    pagesize = 10000
  )

  close(con_in)
  close(con_out)

  batch_reviews <- stream_in(file(batch_file))
  all_friend_reviews[[i]] <- batch_reviews
}
```

```
  unlink(batch_file)
}
```

```
friends_reviews <- bind_rows(all_friend_reviews)
head(friends_reviews) #759580

friends_business_id <- friends_reviews$business_id

business_file <- paste0(data_dir0, "/1.yelp_academic_dataset_business.json")
business <- stream_in(file(business_file), verbose = FALSE)
dim(business)#150346

friend_business<- left_join(
  friends_reviews,business,by = "business_id",
  suffix = c("_user", "_business"))

head(friend_business) #759580
```

```
restaurant_keywords <- c("restaurants", "food","bars")

is_restaurant <- function(categories) {
  if(is.na(categories)) return(FALSE)
  categories_lower <- tolower(categories)
  any(apply(restaurant_keywords, function(kw) grepl(kw, categories_lower)))
}

friend_restaurants <- friend_business |>
  rowwise() |>
  filter(is_restaurant(categories)) |>
  filter(!is.na(attributes$RestaurantsPriceRange2)) |>
  filter(attributes$RestaurantsPriceRange2 != "None") %>%
  mutate(
    price_level = as.numeric(attributes$RestaurantsPriceRange2)
  ) |>
  ungroup()

# Restaurants Data (0327)
friend_restaurants2<-friend_restaurants |> select(-c(review_id,address,postal_code,latitude))
dim(friend_restaurants2) #582780
head(friend_restaurants2)
```

```
#save JSON
write_lines(
  toJSON(friends_reviews, auto_unbox = TRUE),
  "omnivore_friend_reviews_data0319.json"
)
#save JSON
write_lines(
  toJSON(friend_restaurants2, auto_unbox = TRUE),
  "friend_restaurants0319.json"
)

#write_csv(philly_restaurants2, paste0(data_dir, 'philly_restaurants.csv'))
```